

STATEMENT OF WORK

SQUID Magnetometer Upgrade

I. BACKGROUND INFORMATION

The Spin Electronics Group of the Physical Measurement Laboratory requires an upgrade of a SQUID (Superconducting Quantum Interference Device) Magnetometer to a closed cryostat that does not require the frequent (weekly) use of liquid helium (LHe) for use. The present system requires roughly 100 L of LHe weekly to remain functional. The upgrade must not degrade the present technical capabilities of the apparatus and must maintain the ability to purchase preventative maintenance contracts for the entire measurement in the future. The controller must be able to control our current stages and perform closed loop positioning using the resistive encoders that our stages have.

II. PURPOSE

The required work will provide the upgrade in-place without the need for the present unit to be shipped offsite. The unit needs to be upgraded onsite while maintaining present technical capabilities (e.g. sensitivity, temperature ranges, etc.) but will no longer need LHe for operation.

PLACE OF PERFORMANCE

Boulder, CO Building 2 Room 1233 (and potentially adjacent laboratories)

Regular Business Hours 9:00 AM- 5:00 PM during regular business day and non-federal holidays

PERIOD OF PERFORMANCE

12 months ARO

III. SPECIFIC REQUIREMENTS

The upgrade must meet the following specifications:

Must upgrade the system without need to ship the unit offsite and not degrade present technical specification of the unit.

- Maximum ramp rate: 700 Oe/sec
- Temperature Range: 1.8 K – 400 K
- Company must provide electrical and cooling requirements 45 working days for federal employees prior to installation
- If onsite representative is not a US citizen, information required for them to get onsite (passport or green card information) must be provided 30 days prior to their arrival at NIST
- Provide closed system cryostat in order to maintain temperature range including all
 - Compressors
 - Cold head and necessary vacuum equipment/containers
 - Pumps
 - Cabling and hoses necessary to integrate additional equipment into the system
 - Specialized equipment required for regular maintenance to be performed (as allowed by warranty)
 - 12 months warranty on all parts and labor
 - (access to appropriate cooling water and electrical infrastructure will be provided by NIST)
 - Hands-on training for a minimum of 5 people chosen by NIST staff member
- -No prototypes, no refurbished equipment, must meet factory acceptance standards

Government Furnished Property
Quantum Design MPM3 SQUID magnetometer

Inspection & Acceptance

The Government Technical Point of Contact (TPOC) will verify temperature range and sensitivity of operation with the service technician.

Performance Tests:

1. Measurements of a sample will be performed ear the highest and lowest temperatures of the cryostats range.
2. Sensitivity and signal to noise of a typical sample (as chosen by NIST) will be compared with the identical sample made prior to the system upgrade.

NIST may choose at its discretion to forego this part of acceptance testing.

A visual inspection of the equipment will be performed by the NIST POC to identify surface defects or any form of indication that any equipment was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the equipment due to defects and/or nonconformance. The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the equipment.

Ownership of the equipment shall transfer to NIST upon acceptance by the Government.

The Government will test, inspect, and accept or reject the equipment within 30 days of the receipt of the equipment unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

REQUIREMENTS FOR CONTRACTOR PERSONNEL

N/A

V. SCHEDULE OF DELIVERABLES

Define all deliverables and indicate (1) the quantity desired; (2) the format in which each deliverable is to be supplied; (3) the delivery date (based on the number of weeks/days); (4) the place of delivery. The context of each deliverable shall be clear. **Keep in mind that every task does not require a deliverable. Deliverables are always tangible items (i.e., report, minutes, etc.)**

Complete the table providing the information for all the tasks identified above.

Deliverable Number	Description	Format	Due Date	Quantity
1	All hardware and software to upgrade NIST's present MPMS 3 system to a closed cycled cryostat	Hardware and software	12 months ARO	1